Hydrogen Atom

In this note, I will teach my self to derive wavefunction and eigen value from Schrodinger equation of Hydrogen atom.

First, we assume the electron arrounds the nuclear as like the planets arround the sun [Bohr's atom theory] and we assume the motion of electron follows the spherical Coordinat, so schrodinger equation can be expresied for steady state

$$H\psi(r,\theta,\phi) = E\psi(r,\theta,\phi)$$

$$\frac{-\hbar^2}{2m}\nabla^2\psi(r,\theta,\phi) - \frac{Ze^2}{4\pi\varepsilon_0 r}\psi(r,\theta,\phi) = E\psi(r,\theta,\phi)$$

where ∇^2 follows spherical Coordinates (we don't need to derive this Nabla operator)

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\phi^2} \quad \cdots (1)$$

Substitute (1) to schrodinger equation

$$\frac{-\hbar^2}{2\mu}\left(\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right)\psi(r,\theta,\phi) - \frac{Ze^2}{4\pi\varepsilon_0 r}\psi(r,\theta,\phi) = E\psi(r,\theta,\phi)$$

We assume $\psi(r,\theta,\phi)$ is separable by the expression $\psi(r,\theta,\phi) = R(r)\Theta(\theta)\Phi(\phi)$

and $Y_{lm}(\theta,\phi) = \Theta(\theta)\Phi(\phi)$

We can simply the expression

$$\frac{-\hbar^2}{2\mu}\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right]R\Theta\Phi - \frac{Ze^2}{4\pi\varepsilon_0 r}R\Theta\Phi = ER\Theta\Phi$$

$$\Theta\Phi\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}R\right) + R\Phi\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\Theta}{\partial\theta}\right) + R\Theta\frac{1}{\sin^2\theta}\frac{\partial^2\Phi}{\partial\phi^2} - \frac{Ze^2 r}{4\pi\varepsilon_0}R\Theta\Phi = Er^2R\Theta\Phi$$

$$\frac{1}{R}\frac{\partial}{\partial r}\left(r^2\frac{\partial R}{\partial r}\right) + \frac{1}{\Theta}\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\Theta}{\partial\theta}\right) + \frac{1}{\Phi}\frac{1}{\sin^2\theta}\frac{\partial^2\Phi}{\partial\phi^2} + \frac{2\mu}{\hbar^2}\left[\frac{Ze^2 r}{4\pi\varepsilon_0} + Er^2\right] = 0$$

$$\frac{1}{R}\frac{d}{dR}\left(r^2\frac{dR}{dr}\right) + \frac{2\mu}{\hbar^2}\left(\frac{Ze^2 r}{4\pi\varepsilon_0} + Er^2\right) + \frac{1}{\sin\theta}\frac{1}{\Theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right) + \frac{1}{\sin^2\theta}\frac{1}{\Phi}\frac{d^2\Phi}{d\phi^2} = 0$$

We focus to solve Θ and Φ part

$$\frac{1}{\Theta}\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right)+\frac{1}{\Phi}\frac{1}{\sin^2\theta}\frac{d^2\Phi}{d\varphi^2}=k_1 \quad , \text{ where } k_1=-l(l+1)$$

$$\frac{1}{\Theta}\sin\theta\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right)+\frac{1}{\Phi}\frac{d^2\Phi}{d\varphi^2}+l(l+1)\sin^2\theta=0$$

$\frac{1}{\Phi}\frac{d^2\Phi}{d\varphi^2}=-m^2$ and we have $\frac{d^2\Phi}{d\varphi^2}+m^2\Phi=0$, so we have $\Phi\propto e^{\pm im\phi}$

Substitute these solution to angular part

$$\frac{1}{\Theta}\sin\theta\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right)+\left(l(l+1)\sin^2\theta-m^2\right)=0$$

$$\frac{1}{\Theta}\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right)+\left(l(l+1)-\frac{m^2}{\sin^2\theta}\right)=0$$

$$\frac{1}{\Theta}\frac{1}{\sin\theta}\left(\cos\theta\frac{d\Theta}{d\theta}+\sin\theta\frac{d^2\Theta}{d\theta^2}\right)+\left(l(l+1)-\frac{m^2}{\sin^2\theta}\right)=0$$

$$\cot\theta\frac{d\Theta}{d\theta}+\frac{d^2\Theta}{d\theta^2}+\left(l(l+1)-\frac{m^2}{\sin^2\theta}\right)\Theta=0$$

$$\frac{d^2\Theta}{d\theta^2}+\cot\theta\frac{d\Theta}{d\theta}+\left(l(l+1)-\frac{m^2}{\sin^2\theta}\right)\Theta=0$$

We have $x=\cos\theta \Rightarrow \frac{dx}{d\theta}=-\sin\theta$

$$\frac{d}{d\theta}=\frac{d}{dx}\frac{dx}{d\theta}=-\sin\theta\frac{d}{dx}$$

$$\frac{d^2}{d\theta^2}=\frac{d}{d\theta}\left(-\sin\theta\frac{d}{dx}\right)=-\cos\theta\frac{d}{dx}-\sin\theta\frac{d}{d\theta}\frac{d}{dx}=-\cos\theta\frac{d}{dx}-\sin\theta\frac{d^2}{dx^2}\frac{dx}{d\theta}$$

$$=-\cos\theta\frac{d}{dx}+\sin^2\theta\frac{d^2}{dx^2}$$

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$$\sin^2\theta\frac{d^2\Theta}{dx^2}-\cos\theta\frac{d\Theta}{dx}+\frac{\cos\theta}{\sin\theta}\left(-\sin\theta\frac{d\Theta}{dx}\right)+\left(l(l+1)-\frac{m^2}{\sin^2\theta}\right)\Theta=0$$

$$\sin^2\theta\frac{d^2\Theta}{dx^2}-2\cos\theta\frac{d\Theta}{dx}+\left(l(l+1)-\frac{m^2}{\sin^2\theta}\right)\Theta=0$$

$$(1-x^2)\frac{d^2\Theta}{dx^2}-2x\frac{d\Theta}{dx}+\left(l(l+1)-\frac{m^2}{(1-x^2)}\right)\Theta=0$$

the solution for this equation is

$$\boxed{Y_{lm}(\theta,\phi)=(-1)^m\sqrt{\frac{(2l+1)}{4\pi}\frac{(l-m)!}{(l+m)!}}\,P_{lm}(\cos\theta)\,e^{im\phi}}$$

Now we focus to Radial part

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)+\frac{2\mu}{\hbar^2}\left[\frac{Ze^2}{4\pi\varepsilon_0 r}+E-\frac{l(l+1)}{r^2}\right]R=0$$

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)+\left(\frac{2\mu E}{\hbar^2}+\frac{Ze^2}{2\pi\varepsilon_0\hbar^2 r}-\frac{l(l+1)}{r^2}\right)R=0$$

$$\frac{1}{r^2}\left[2r\frac{dR}{dr}+r^2\frac{d^2R}{dr^2}\right)+\left(\frac{2\mu E}{\hbar^2}+\frac{Ze^2}{2\pi\varepsilon_0\hbar^2 r}-\frac{l(l+1)}{r^2}\right)R=0$$

$$R''+\frac{2}{r}R'+\left(\frac{2\mu E}{\hbar^2}+\frac{Ze^2}{2\pi\varepsilon_0\hbar^2 r}-\frac{l(l+1)}{r^2}\right]R=0$$

untuk memudahkan, misalnya $u=rR \Rightarrow R=\frac{u}{r} \Rightarrow \frac{dR}{dr}=-\frac{1}{r^2}u+\frac{1}{r}\frac{du}{dr}$

$$\frac{d^2R}{dr^2}=\frac{2}{r^3}u-\frac{1}{r^2}u'-\frac{1}{r^2}u'+\frac{1}{r}u''=\frac{1}{r}u''-\frac{2}{r^2}u'+\frac{2}{r^3}u$$

$$\frac{1}{r}u''-\frac{2}{r^2}u'+\frac{2}{r^3}u+\frac{2}{r}\left[-\frac{1}{r^2}u+\frac{1}{r}u'\right]+\left[\frac{2\mu E}{\hbar^2}+\frac{Ze^2}{2\pi\varepsilon_0\hbar^2 r}-\frac{l(l+1)}{r^2}\right]\frac{u}{r}=0$$

$$\frac{1}{\cancel{r}}u''+\left[\frac{2\mu E}{\hbar^2}+\frac{Ze^2}{2\pi\varepsilon_0\hbar^2 r}-\frac{l(l+1)}{r^2}\right]\frac{u}{\cancel{r}}=0$$

$$\boxed{u''+\left[\frac{2\mu E}{\hbar^2}+\frac{Ze^2}{2\pi\varepsilon_0\hbar^2 r}-\frac{2\mu l(l+1)}{\hbar^2 r^2}\right]u=0}$$

We know for bound electron $E<0$ and $k^2=\frac{-2\mu E}{\hbar^2}$ and

$$u''+\left(-k^2+\frac{2}{a_0 r}-\frac{l(l+1)}{r^2}\right)u=0$$

We define $\rho=2kr \Rightarrow \frac{d}{dr}=\frac{d}{d\rho}\frac{d\rho}{dr}=2k\frac{d}{d\rho}$, $\frac{d^2}{dr^2}=2k\frac{d^2}{d\rho^2}\frac{d\rho}{dr}=4k^2\frac{d^2}{d\rho^2}$

$$4k^2\frac{d^2u}{d\rho^2}+\left[-k^2+\frac{4k}{\rho a_0}-\frac{4k^2 l(l+1)}{\rho^2}\right]u=0$$

$$\frac{d^2u}{d\rho^2}+\left[-\frac{1}{4}+\frac{1}{ka_0\rho}-\frac{l(l+1)}{\rho^2}\right]u=0$$

for $r \to 0 \to \frac{1}{\rho^2} \gg \frac{1}{\rho}$

$$\frac{d^2u}{d\rho^2} + \left[-\frac{1}{4} + \frac{1}{ka_0\rho} - \frac{l(l+1)}{\rho^2}\right]u = 0$$

$$u'' - \frac{l(l+1)}{\rho^2}u = 0 \Rightarrow u \propto \rho^s,\ u' = s\rho^{s-1},\ u'' = s(s-1)\rho^{s-2}$$

$$\sum_{s=2} s(s-1)\rho^{s-2} - l(l+1)\rho^{-2}\rho^s = 0$$

$$s^2 - s - l^2 - l = 0$$

$$s^2 - l^2 - (s+l) = 0$$

$$(s+l)(s-l) - (s+l) = 0$$

$$(s+l)(s-l-1) = 0$$

we have $s = -l$ or $s = l+1$

so $\quad u(\rho) \propto A\rho^{-l} + B\rho^{l+1}$

when $r \to 0$, we have solution $u(\rho) \propto B\rho^{l+1}$

and when $r \to \infty$ so $\rho \to \infty$ so we have

$$u'' - \frac{1}{4}u = 0 \Rightarrow \left(\frac{d^2}{d\rho^2} - \frac{1}{4}\right)u = 0$$

$D = \pm\frac{1}{2}$, $u \propto e^{\pm \rho/2}$, so we guess our wavefunction

$$u(\rho) = \rho^{l+1} e^{\pm \rho/2} f(\rho)$$

Substitute there function to our Schrodinger equation

$$u'' + \left[-\frac{1}{4} + \frac{1}{ka_0\rho} + \frac{l(l+1)}{\rho^2}\right]u = 0$$

$$u' = \left[\frac{(l+1)}{\rho} - \frac{1}{2}\right]\rho^{l+1}e^{-\rho/2}f + \rho^{l+1}e^{-\rho/2}f'$$

$$u'' = \left[\left[\frac{(l+1)}{\rho} - \frac{1}{2}\right]\left[\frac{l+1}{\rho} - \frac{1}{2}\right] - \frac{(l+1)}{\rho^2}\right]\rho^{l+1}e^{-\rho/2}f + \left[\frac{(l+1)}{\rho} - \frac{1}{2}\right]\rho^{l+1}e^{-\rho/2}f'$$

$$+ \left[\frac{(l+1)}{\rho} - \frac{1}{2}\right]e^{-\rho/2}\rho^{l+1}f' + \rho^{l+1}e^{-\rho/2}f''$$

$$u'' = \left[\left[\left(\frac{l+1}{\rho} - \frac{1}{2}\right)\right]^2 - \frac{(l+1)}{\rho^2}\right]f + 2\left[\left(\frac{l+1}{\rho} - \frac{1}{2}\right)f' + f''\right]e^{-\rho/2}\rho^{l+1}$$

$$u'' = \left[\left[\left(\frac{(\ell+1)}{\rho} - \frac{1}{2}\right]^2 - \frac{(\ell+1)}{\rho^2}\right] f + 2\left[\frac{(\ell+1)}{\rho} - \frac{1}{2}\right) f' + f''\right] e^{-\rho/2} \rho^{l+1}$$

$$f'' + 2\left[\frac{(\ell+1)}{\rho} - \frac{1}{2}\right] f' + \left[\left(\frac{(\ell+1)}{\rho} - \frac{1}{2}\right]^2 - \cancel{\frac{(\ell+1)}{\rho^2}}\right] f + \left[-\frac{1}{4} + \frac{1}{ka_0\rho} + \cancel{\frac{\ell(\ell+1)}{\rho^2}}\right] f = 0$$

$$f'' + 2\left[\frac{(\ell+1)}{\rho} - \frac{1}{2}\right] f' + \left[\frac{(\ell+1)^2}{\rho^2} - \frac{(\ell+1)}{\rho} + \frac{1}{4} - \frac{1}{4} + \frac{1}{ka_0\rho}\right] f = 0$$

$$f'' + 2\left[\frac{(\ell+1)}{\rho} - \frac{1}{2}\right] f' + \left[\left(\frac{(\ell+1)}{\rho}\left[\left(\frac{\ell+1}{\rho}\right) - 1\right] + \frac{1}{ka_0\rho}\right] f = 0$$

for $r \to \infty$

$$f'' + 2\left[\frac{(\ell+1)}{\rho} - \frac{1}{2}\right] f' + \left[\frac{1}{ka_0} - (\ell+1)\right] \frac{f}{\rho} = 0$$

we define $f = \sum_{n=0} a_n \rho^n \Rightarrow f'' = \sum_{n=2} n(n-1) a_n \rho^{n-2}$ and $f' = \sum_{n=1} n a_n \rho^{n-1}$

$$\sum_{n=2} n(n-1) a_n \rho^{n-2} + 2\sum_{n=1}\left(\frac{(\ell+1)}{\rho} - \frac{1}{2}\right) a_n \rho^{n-1} + \left[\frac{1}{ka_0} - (\ell+1)\right] \sum_{n=0} a_n \rho^{n-1} = 0$$

$$\sum_{n=0} (n+2)(n+1) a_{n+2} \rho^n + 2\sum_{n=0}\left(\frac{(\ell+1)}{\rho} - \frac{1}{2}\right] a_{n+1} \rho^n + \left(\frac{1}{ka_0} - (\ell+1)\right) \sum_{n=0} a_n \rho^{n-1} = 0$$